<u>DNA Metabarcoding data processing and analysis pipeline:</u>

Seb-Qiime2-Scripts:

#login
ssh ntq237@epyc.simcenter.utc.edu

#activate qiime2
srun --time=1-00:00:00 --partition=general --ntasks=1 --pty /bin/bash -l
module load anaconda
source activate qiime2-amplicon-2025.10

To logout:
logout (2x)
---
*may not include .qza→.qzv or export codes; use discretion (e.g., view outputs between steps to check for errors)

Example export codes:

qiime tools export --input-path rep-seqs.qza --output-path exports

qiime tools export --input-path sv-table.qza --output-path exports → biom convert -i feature-table.biom -o rarefied-feature-table.tsv --to-tsv

qiime tools export --input-path classification-blast.qza --output-path exports

qiime tools export --input-path search-results.qza --output-path exports

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/florida/rep-seqs.qzv Desktop/exports/

Upload files to qiime (logged out in terminal) into: new directory → raw_reads:
scp -r Desktop/miami25-w-raw-reads/*.fastq.gz
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-w/raw-reads/

scp -r Desktop/miami25-s-raw-reads/*.fastq.gz
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-s/raw-reads/

Importing raw reads to qiime2:

#Summary: In the directory the file is located, give the file execution permissions → run the script – specify pathway to the “raw_reads” directory with all your sequencing files: <chmod +x metabarcoding-trim-fastp.sh> and <./metabarcoding-trim-fastp.sh raw_reads/>

Make a manifest file to upload to Qiime2:

#Copy the following script into nano and make a file: Name it “mkmanifest.py”

```
#!/usr/bin/env python3

import argparse
import os
import glob

def parse_args():
    parser = argparse.ArgumentParser(description="Generate QIIME2 manifest from paired-end FASTQ files.")
    parser.add_argument(
        "-i", "--input-dir", required=True,
        help="Directory containing paired-end FASTQ files with *_R1_* in the filename."
    )
    parser.add_argument(
        "-o", "--output", default="manifest.tsv",
        help="Output manifest filename (default: manifest.tsv)"
    )
    return parser.parse_args()

def main():
    args = parse_args()
    input_dir = os.path.abspath(args.input_dir)
    r1_files = sorted(glob.glob(os.path.join(input_dir, "*_R1_*")))

    if not r1_files:
        raise FileNotFoundError("No *_R1_* files found in the specified directory.")

    with open(args.output, "w") as out:
        out.write("sample-id\tforward-absolute-filepath\treverse-absolute-filepath\n")
        for r1_path in r1_files:
            r2_path = r1_path.replace("_R1_", "_R2_")
            if not os.path.exists(r2_path):
                print(f"Warning: reverse read not found for {r1_path}, skipping.")
                continue

            basename = os.path.basename(r1_path)
            sample_id = basename.split("_")[0]
            out.write(f"{sample_id}\t{r1_path}\t{r2_path}\n")

if __name__ == "__main__":
    main()
```

#Run the script
chmod +x mkmanifest.py
./mkmanifest.py -i raw-reads/

Importing manifest file and refining reads:

```
qiime tools import \
--type 'SampleData[PairedEndSequencesWithQuality]' \
--input-path manifest.tsv \
--input-format PairedEndFastqManifestPhred33V2 \
--output-path demux-paired-end.qza

qiime demux summarize \
--i-data demux-paired-end.qza \
--o-visualization demux-paired-end.qzv
```

View output:

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-w/demux-paired-end.qzv
Desktop/exports/

~~Trim adapters + primers using Cutadapt:~~

```
qiime cutadapt trim-paired \
--i-demultiplexed-sequences demux-paired-end.qza \
--p-adapter-f GTAGGTGAACCTGCAGAAGGATCA \
--p-adapter-r GACGGGCGGTGTGTAC \
--p-match-read-wildcards \
--p-match-adapter-wildcards \
--verbose \
--o-trimmed-sequences demux-paired-end-trimmed-1.qza \

qiime cutadapt trim-paired \
--i-demultiplexed-sequences demux-paired-end-trimmed-1.qza \
--p-front-f GTACACACCGCCCGTC \
--p-front-r TGATCCTTCTGCAGGTTCACCTAC \
--p-match-read-wildcards \
--p-match-adapter-wildcards \
--verbose \
--o-trimmed-sequences demux-paired-end-trimmed-2.qza

qiime demux summarize \
--i-data demux-paired-end-trimmed-2.qza \
--o-visualization demux-paired-end-trimmed-2.qzv

scp -r
ntq237r@epyc.simcenter.utc.edu:/home/ntq237r/miami25/demux-paired-end-trimmed-2.qzv
Desktop/exports/
```

~~***Compare none discarded vs. untrimmed discarded to see how much is lost. Discuss which one to use with Francesca.***~~

```
qiime cutadapt trim-paired \
--i-demultiplexed-sequences demux-paired-end.qza \
--p-adapter-f GTAGGTGAACCTGCAGAAGGATCA \
--p-adapter-r GACGGGCGGTGTGTAC \
--p-match-read-wildcards \
--p-match-adapter-wildcards \
--p-discard-untrimmed \
--verbose \
--o-trimmed-sequences demux-paired-end-trimmed-du-1.qza \

qiime cutadapt trim-paired \
--i-demultiplexed-sequences demux-paired-end-trimmed-du-1.qza \
--p-front-f GTACACACCGCCCGTC \
--p-front-r TGATCCTTCTGCAGGTTCACCTAC \
--p-match-read-wildcards \
--p-match-adapter-wildcards \
--p-discard-untrimmed
--verbose \
--o-trimmed-sequences demux-paired-end-trimmed-du-2.qza
```

**Remember to modify Dada2 outputs so they don't replace the ones without the discard parameter!!!

```
qiime dada2 denoise-paired \
--i-demultiplexed-seqs demux-paired-end-trimmed-du-2.qza \
--o-table sv-table-du.qza \
--o-representative-sequences rep-seqs-du.qza \
--p-n-threads 30 \
--p-trim-left-f 0 \
--p-trim-left-r 0 \
--p-trunc-len-f 0 \
--p-trunc-len-r 0 \
--o-denoising-stats stats-dada2-du.qza
```

~~qiime dada2 denoise-paired \~~
~~--i-demultiplexed-seqs demux-paired-end-trimmed-2.qza \~~
~~--o-table sv-table.qza \~~
~~--o-representative-sequences rep-seqs.qza \~~
~~--p-n-threads 30 \~~
~~--p-trim-left-f 0 \~~

~~--p-trim-left-r 0 \~~
~~--p-trunc-len-f 0 \~~
~~--p-trunc-len-r 0 \~~
~~--o-denoising-stats stats-dada2.qza~~

If you use Cutadapt, please cite DOI:10.14806/ej.17.1.200 .

Denoise (tweak nucleotides cut off on each end based on length + quality)
#Takes 6+ hours, so make a nano file (based on *SLURM example dada2.slurm* with #SBATCH parameters found below) → name “dada2.slurm” → type “sbatch dada2.slurm” to submit a SLURM job in the background.

```
qiime dada2 denoise-paired \
--i-demultiplexed-seqs paired-end-demux.qza \
--o-denoising-stats dada2-stats.qza \
--o-table sv-table.qza \
--o-representative-sequences rep-seqs.qza \
--p-n-threads 30 \
--p-trim-left-f 16 \
--p-trim-left-r 24 \
--p-trunc-len-f 100 \
--p-trunc-len-r 100
```

*NOTE: If submitting a SLURM job, make sure “cpus-per-task” = “--p-n-threads”

SLURM example dada2.slurm:

```
#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=30
#SBATCH --mail-type=END
#SBATCH --mail-user=ntq237@mocs.utc.edu

qiime dada2 denoise-paired \
--i-demultiplexed-seqs paired-end-demux.qza \
--o-denoising-stats dada2-stats.qza \
--o-table sv-table.qza \
--o-representative-sequences rep-seqs.qza \
--p-n-threads 30 \
--p-trim-left-f 16 \
--p-trim-left-r 24 \
--p-trunc-len-f 100 \
--p-trunc-len-r 100
```

Download sv-table, rep-seqs, and stats-dada2 files

```
qiime feature-table summarize \
--i-table sv-table-du.qza \
--o-visualization sv-table-du.qzv

qiime feature-table tabulate-seqs \
--i-data rep-seqs-du.qza \
--o-visualization rep-seqs-du.qzv
```

**cd to exports and check that it is empty with* the cmd: *ls – DO NOT GET YOUR FILES MIXED UP WITH OLD ONES*

```
qiime tools export --input-path stats-dada2-du.qza --output-path exports
```

View output:

```
scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-s/rep-seqs-du.qzv Desktop/exports/
scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-s/sv-table-du.qzv Desktop/exports/
scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-s/exports/stats.tsv Desktop/exports/
```

Filtering based on sequence length: *dl = discarded by length, rl = retained by length

```
qiime rescript filter-seqs-length \
  --i-sequences rep-seqs-du.qza \
  --p-global-min 96 \
  --p-global-max 260 \
  --o-discarded-seqs rep-seqs-dl.qza \
  --o-filtered-seqs rep-seqs-rl.qza

qiime feature-table filter-features \
  --i-table sv-table-du.qza \
  --m-metadata-file rep-seqs-rl.qza \
  --o-filtered-table sv-table-rl.qza

qiime feature-table tabulate-seqs \
--i-data rep-seqs-rl.qza \
--o-visualization rep-seqs-rl.qzv

qiime feature-table summarize \
--i-table sv-table-rl.qza \
--o-visualization sv-table-rl.qzv
```

```
scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-s/rep-seqs-rl.qzv
Desktop/exports/
scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-s/sv-table-rl.qzv
Desktop/exports/
```

Frequency filter out bad reads (note excluded samples for resequencing) *ff = frequency filtered

```
qiime feature-table filter-samples \
--i-table sv-table-rl.qza \
--p-min-frequency 0 \
--o-filtered-table sv-table-rl-ff.qza
```

View the filtered table to confirm it is correct: .qza → .qvz

```
qiime feature-table summarize \
--i-table sv-table-rl-ff.qza \
--o-visualization sv-table-rl-ff.qzv
```

View output

Filter rep-seqs.qza based on the frequency-filtered-table.qza

```
qiime feature-table filter-seqs \
--i-data rep-seqs-rl.qza \
--i-table sv-table-rl-ff.qza \
--o-filtered-data rep-seqs-rl-ff.qza
```

View output

```
qiime feature-table tabulate-seqs \
--i-data rep-seqs-rl-ff.qza \
--o-visualization rep-seqs-rl-ff.qzv
```

Combine water and sediment sample data (if necessary)

```
qiime feature-table merge-seqs \
--i-data rep-seqs-rl-w.qza rep-seqs-rl-s.qza \
--o-merged-data rep-seqs-rl-all.qza

qiime feature-table merge \
--i-tables sv-table-rl-w.qza sv-table-rl-s.qza \
--o-merged-table sv-table-rl-all.qza
```

View outputs

```
qiime feature-table tabulate-seqs \
--i-data rep-seqs-rl-all.qza \
```

```
--o-visualization rep-seqs-rl-all.qzv

qiime feature-table summarize \
--i-table sv-table-rl-all.qza \
--o-visualization sv-table-rl-all.qzv

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/rep-seqs-rl-all.qzv
Desktop/exports/

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/sv-table-rl-all.qzv
Desktop/exports/
```

Classify sequences using consensus BLAST

```
#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=15

qiime feature-classifier classify-consensus-blast \
--i-query rep-seqs.qza \
--i-reference-taxonomy ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-tax.qza \
--i-reference-reads ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-dna-derep-seqs.qza \
--o-classification classification-blast.qza \
--o-search-results search-results.qza \
--p-perc-identity 0.90 \
--p-query-cov 0.8 \
--p-maxaccepts 1
```

View classification

```
qiime tools export --input-path classification-blast.qza --output-path exports

qiime tools export --input-path blast-search-results.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/taxonomy.tsv
Desktop/exports/

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/blast6.tsv
Desktop/exports/
```

#Or you can make a .qzv instead if you prefer
qiime metadata tabulate \
--m-input-file classification-blast.qza \
--o-visualization classification-blast.qzv

View output

Filter out non-meiofauna from sv-table and rep-seqs using assigned taxonomies to ASVs(broad first-pass to simplify further exclusions): f1 = “filtered once”
qiime taxa filter-table \
--i-table sv-table-rl-all.qza \
--i-taxonomy classification-blast.qza \
--p-include Animalia \
--p-exclude
Acanthocephala,Vertebrata,Tunicata,Glenodinium,Cestoda,Monogenea,Trematoda,Bivalvia,Sanchytrium,Insecta,Ellipura,Araneae,Diplopoda,Myxozoa,Anthozoa,Ctenophora,Echinodermata,Cephalopoda,Polyplacophora,Opiliones,Scorpiones,Rhynchobdellida \
--o-filtered-table f1-sv-table.qza

qiime taxa filter-seqs \
--i-sequences rep-seqs-rl-all.qza \
--i-taxonomy classification-blast.qza \
--p-include Animalia \
--p-exclude
Acanthocephala,Vertebrata,Tunicata,Glenodinium,Cestoda,Monogenea,Trematoda,Bivalvia,Sanchytrium,Insecta,Ellipura,Araneae,Diplopoda,Myxozoa,Anthozoa,Ctenophora,Echinodermata,Cephalopoda,Polyplacophora,Opiliones,Scorpiones,Rhynchobdellida \
--o-filtered-sequences f1-rep-seqs.qza

Export sequences
qiime feature-table summarize \
--i-table f1-sv-table.qza \
--o-visualization f1-sv-table.qzv

qiime feature-table tabulate-seqs \
--i-data f1-rep-seqs.qza \
--o-visualization f1-rep-seqs.qzv

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f1-sv-table.qzv Desktop/exports

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f1-rep-seqs.qzv Desktop/exports

Reclassify with BLAST using the filtered rep-seqs; download search-output file for each!

```
#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=8
#SBATCH --mail-type=END
#SBATCH --mail-user=ntq237@mocs.utc.edu

qiime feature-classifier classify-consensus-blast \
--i-query f1-rep-seqs.qza \
--i-reference-taxonomy ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-tax.qza \
--i-reference-reads ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-dna-derep-seqs.qza \
--o-classification f1-classification-blast.qza \
--o-search-results f1-search-results.qza \
--p-perc-identity 0.90 \
--p-maxaccepts 1
```

View output

```
qiime tools export --input-path f1-classification-blast.qza --output-path exports

qiime tools export --input-path f1-search-results.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/taxonomy.tsv
Desktop/exports/

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/blast6.tsv
Desktop/exports/
```

Remove controls: Make a metadata file (excel) with a list of samples to keep in a column

```
scp -r Desktop/ca-miami-all/samples-to-keep.csv
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/

qiime feature-table filter-samples \
  --i-table f1-sv-table.qza \
  --m-metadata-file samples-to-keep.csv \
  --o-filtered-table f2-sv-table.qza
```

```
qiime feature-table filter-seqs \
--i-data f1-rep-seqs.qza \
--i-table f2-sv-table.qza \
--o-filtered-data f2-rep-seqs.qza

qiime feature-table summarize \
--i-table f2-sv-table.qza \
--o-visualization f2-sv-table.qzv

qiime feature-table tabulate-seqs \
--i-data f2-rep-seqs.qza \
--o-visualization f2-rep-seqs.qzv

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f2-sv-table.qzv
Desktop/exports

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f2-rep-seqs.qzv
Desktop/exports

#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=8
#SBATCH --mail-type=END
#SBATCH --mail-user=ntq237@mocs.utc.edu

qiime feature-classifier classify-consensus-blast \
--i-query f2-rep-seqs.qza \
--i-reference-taxonomy ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-tax.qza \
--i-reference-reads ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-dna-derep-seqs.qza \
--o-classification f2-classification-blast.qza \
--o-search-results f2-search-results.qza \
--p-perc-identity 0.90 \
--p-maxaccepts 1
```

View output

```
qiime tools export --input-path f2-classification-blast.qza --output-path exports

qiime tools export --input-path f2-search-results.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/taxonomy.tsv
Desktop/exports/

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/blast6.tsv
Desktop/exports/
```

Remove ASVs present in less than 3 samples

```
qiime feature-table filter-features \
  --i-table f2-sv-table.qza \
  --p-min-samples 3 \
  --o-filtered-table f3-sv-table.qza

qiime feature-table filter-seqs \
--i-data f2-rep-seqs.qza \
--i-table f3-sv-table.qza \
--o-filtered-data f3-rep-seqs.qza

qiime feature-table summarize \
--i-table f3-sv-table.qza \
--o-visualization f3-sv-table.qzv

qiime feature-table tabulate-seqs \
--i-data f3-rep-seqs.qza \
--o-visualization f3-rep-seqs.qzv

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f3-sv-table.qzv
Desktop/exports

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f3-rep-seqs.qzv
Desktop/exports
```

Make a file that contains the ASVs filtered out by excluding controls (based on f1 - meiofauna only)

```
qiime tools export --input-path f1-sv-table.qza --output-path original
qiime tools export --input-path f2-sv-table.qza --output-path filtered
```

```
biom convert -i original/feature-table.biom -o original.tsv --to-tsv
biom convert -i filtered/feature-table.biom -o filtered.tsv --to-tsv

cut -f1 original.tsv | tail -n +2 | sort > original-ids.txt
cut -f1 filtered.tsv | tail -n +2 | sort > filtered-ids.txt

comm -23 original-ids.txt filtered-ids.txt > control-asvs.txt

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/control-asvs.txt
Desktop/exports/
```

Make a set of sv-table + seqs files that only contains the controls

```
scp -r Desktop/ca-miami-all/controls.csv
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/

qiime feature-table filter-samples \
  --i-table f1-sv-table.qza \
  --m-metadata-file controls.csv \
  --o-filtered-table control-sv-table.qza


qiime feature-table filter-seqs \
--i-data f1-rep-seqs.qza \
--i-table control-sv-table.qza \
--o-filtered-data control-rep-seqs.qza

qiime feature-table summarize \
--i-table control-sv-table.qza \
--o-visualization control-sv-table.qzv

qiime feature-table tabulate-seqs \
--i-data control-rep-seqs.qza \
--o-visualization control-rep-seqs.qzv

qiime tools export --input-path control-sv-table.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/control-sv-table.qzv
Desktop/exports
```

```
scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/control-rep-seqs.qzv
Desktop/exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/rarefied-feature-t
able.tsv Desktop/exports

#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=8
#SBATCH --mail-type=END
#SBATCH --mail-user=ntq237@mocs.utc.edu

qiime feature-classifier classify-consensus-blast \
--i-query control-rep-seqs.qza \
--i-reference-taxonomy ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-tax.qza \
--i-reference-reads ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-dna-derep-seqs.qza \
--o-classification control-classification-blast.qza \
--o-search-results control-search-results.qza \
--p-perc-identity 0.90 \
--p-maxaccepts 1
```

View output

```
qiime tools export --input-path control-classification-blast.qza --output-path exports

qiime tools export --input-path control-search-results.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/taxonomy.tsv
Desktop/exports/

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/blast6.tsv
Desktop/exports/
```

Make a set of files: 1 that contains the PCR controls, and another that contains the bench controls

```
scp -r Desktop/ca-miami-all/c2.csv
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/controls/

qiime feature-table filter-samples \
  --i-table f1-sv-table.qza \
  --m-metadata-file c2.csv \
  --o-filtered-table c2-sv-table.qza


qiime feature-table filter-seqs \
--i-data f1-rep-seqs.qza \
--i-table c2-sv-table.qza \
--o-filtered-data c2-rep-seqs.qza

qiime feature-table summarize \
--i-table c2-sv-table.qza \
--o-visualization c2-sv-table.qzv

qiime feature-table tabulate-seqs \
--i-data c2-rep-seqs.qza \
--o-visualization c2-rep-seqs.qzv

qiime tools export --input-path c1-sv-table.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/controls/c2-sv-table.qzv
Desktop/exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/controls/c2-rep-seqs.qzv
Desktop/exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/rarefied-feature-t
able.tsv Desktop/exports

#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
```

```
#SBATCH --cpus-per-task=8
#SBATCH --mail-type=END
#SBATCH --mail-user=ntq237@mocs.utc.edu

qiime feature-classifier classify-consensus-blast \
--i-query c2-rep-seqs.qza \
--i-reference-taxonomy ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-tax.qza \
--i-reference-reads ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-dna-derep-seqs.qza \
--o-classification c2-classification-blast.qza \
--o-search-results c2-search-results.qza \
--p-perc-identity 0.90 \
--p-maxaccepts 1
```

View output

```
qiime tools export --input-path c2-classification-blast.qza --output-path exports

qiime tools export --input-path c2-search-results.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/controls/exports/taxonomy.tsv Desktop/exports/

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/controls/exports/blast6.tsv Desktop/exports/
```

Filter out samples with < 1000 reads

```
qiime feature-table filter-samples \
--i-table f3-sv-table.qza \
--p-min-frequency 1000 \
--o-filtered-table f4-sv-table.qza

qiime feature-table filter-seqs \
--i-data f3-rep-seqs.qza \
--i-table f4-sv-table.qza \
--o-filtered-data f4-rep-seqs.qza

qiime feature-table summarize \
--i-table f4-sv-table.qza \
--o-visualization f4-sv-table.qzv
```

```
qiime feature-table tabulate-seqs \
--i-data f4-rep-seqs.qza \
--o-visualization f4-rep-seqs.qzv

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f4-sv-table.qzv
Desktop/exports

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f4-rep-seqs.qzv
Desktop/exports
```

Filter out ASVs with < 10 reads

```
qiime feature-table filter-features \
  --i-table f4-sv-table.qza \
  --p-min-frequency 10 \
  --o-filtered-table f5-sv-table.qza


qiime feature-table filter-seqs \
--i-data f4-rep-seqs.qza \
--i-table f5-sv-table.qza \
--o-filtered-data f5-rep-seqs.qza

qiime feature-table summarize \
--i-table f5-sv-table.qza \
--o-visualization f5-sv-table.qzv

qiime feature-table tabulate-seqs \
--i-data f5-rep-seqs.qza \
--o-visualization f5-rep-seqs.qzv

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f5-sv-table.qzv
Desktop/exports

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f5-rep-seqs.qzv
Desktop/exports

#!/bin/bash

#SBATCH --job-name=qiime_job_seb
#SBATCH --output=output.txt
#SBATCH --error=error.txt
#SBATCH --partition=general
#SBATCH --nodes=1
```

```
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=8
#SBATCH --mail-type=END
#SBATCH --mail-user=ntq237@mocs.utc.edu

qiime feature-classifier classify-consensus-blast \
--i-query f5-rep-seqs.qza \
--i-reference-taxonomy ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-tax.qza \
--i-reference-reads ~/databases/Qiime_18S_data/silva-138-ssu-nr99-18s-dna-derep-seqs.qza \
--o-classification f5-classification-blast.qza \
--o-search-results f5-search-results.qza \
--p-perc-identity 0.90 \
--p-maxaccepts 1
```

View output

```
qiime tools export --input-path f5-classification-blast.qza --output-path exports

qiime tools export --input-path f5-search-results.qza --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/taxonomy.tsv
Desktop/exports/

scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/blast6.tsv
Desktop/exports/
```

Get ASVs per sample

```
qiime diversity alpha \
  --i-table f5-sv-table.qza \
  --p-metric observed_features \
  --o-alpha-diversity f5-richness.qza

qiime tools export \
  --input-path f5-richness.qza \
  --output-path exports

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/f5-richness/alpha-diversit
y.tsv Desktop/exports/
```

Biodiversity Analysis:

Generate a neighbor-joining tree using meiofauna-rep-seqs.qza

```
qiime phylogeny align-to-tree-mafft-fasttree \
--i-sequences f5-rep-seqs.qza \
--o-alignment f5-aligned.qza \
--o-masked-alignment f5-masked-aligned.qza \
--o-tree f5-unrooted-tree.qza \
--o-rooted-tree f5-rooted-tree.qza
```

```
scp -r Desktop/ca-miami-all/miami25-mapping-file.txt
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/
```

```
qiime diversity core-metrics-phylogenetic \
--i-phylogeny f5-rooted-tree.qza \
--i-table f5-sv-table.qza \
--p-sampling-depth 2276 \
--m-metadata-file miami25-mapping-file.txt \
--output-dir core-metrics-results
```

```
qiime diversity alpha-group-significance \
--i-alpha-diversity faith_pd_vector.qza \
--m-metadata-file miami25-mapping-file.txt \
--o-visualization faith-pd-group-significance.qzv
```

```
scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/core-metrics-results/faith
-pd-group-significance.qzv Desktop/exports
```

Barplot for illustrative purposes

```
qiime taxa barplot \
--i-table f5-sv-table.qza \
--i-taxonomy f5-classification-blast.qza \
--m-metadata-file miami25-mapping-file.txt \
--o-visualization taxa-bar-plots.qzv
```

```
scp -r ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/taxa-bar-plots.qzv
Desktop/exports
```

#barplot presence/absence: pa = presence/absence

```
qiime feature-table relative-frequency \
```

```
--i-table f5-sv-table.qza \
--o-relative-frequency-table pa-f5-sv-table.qza

qiime taxa barplot \
--i-table pa-f5-sv-table.qza \
--i-taxonomy f5-classification-blast.qza \
--m-metadata-file miami25-mapping-file.tsv \
--o-visualization pa-taxa-bar-plots.qzv
```

Statistical Analyses in R

**generated a rarefied-sv-table is used for adonis (I'm pretty sure it's NOT search results)*

```
qiime tools export --input-path f1-sv-table.qza --output-path exports
biom convert -i feature-table.biom -o rarefied-feature-table.tsv --to-tsv

scp -r
ntq237@epyc.simcenter.utc.edu:/storage/home/ntq237/ca-miami25-all/exports/rarefied-feature-table.tsv Desktop/exports/
```

Using a script in R, replace rep-seq random IDs with Phylum name using the classification.csv
PERMANOVA on species composition data: First we transposed our meiofauna-search-results.csv (containing all the feature IDs) to fit a format that R will use, then we use software in R called adonis to perform the permanova.

this involves making an excel file with the adjusted taxon to include by the IDs, then importing that into the rep_seqs.fasta file

R script: # rename fasta sequences

```
library(phylotools)
fasta<-read.fasta('florida-animalia-bayes-rep-seqs.fasta')
dim(fasta)
retable<-read.csv('florida-animalia-bayes.csv', header=T)
rename.fasta(infile = 'florida-animalia-bayes-rep-seqs.fasta', ref_table=retable, outfile =
 "florida-animalia-bayes-rep-seqs-renamed.fasta")
head(fasta)

library(phylotools)
fasta<-read.fasta('florida-animalia-blast-rep-seqs.fasta')
dim(fasta)
retable<-read.csv('florida-animalia-blast.csv', header=T)
rename.fasta(infile = 'florida-animalia-blast-rep-seqs.fasta', ref_table=retable, outfile =
 "florida-animalia-blast-rep-seqs-renamed.fasta")
head(fasta)
```

R script.To check the best model.

```
# Load necessary package
library(MASS)

# Define multiple regression models. In our case, model1<-lm; model2<-glm; model3<-glm.nb
model1 <- lm(mpg ~ wt + hp, data = data)
model2 <- glm(mpg ~ wt + hp + cyl, data = data)
model3 <- glm.nb(mpg ~ wt + hp + cyl + disp, data = data)

# Compare AIC values
AIC_values <- AIC(model1, model2, model3)
print(AIC_values)

# Identify the best model (lowest AIC)
best_model <- stepAIC(model3, direction = "both", trace = FALSE)
summary(best_model)
```

Classifications can be used to make a barplot: barplot.qvz shows the different levels of taxonomic diversity and measures biodiverity

```
#barplot
qiime taxa barplot   --i-table sv-table-final.qza --i-taxonomy classification-final.qza
--m-metadata-file mapping_file.tsv  --o-visualization taxa-bar-plots.qzv
```

NOTES:

Helpful metabarcoding guide:
https://github.com/Joseph7e/HCGS_Metabarcoding_Tutorials?tab=readme-ov-file

To update Qiime2:

```
#Login to the epyc node and load anaconda
ssh <email login>
srun --time=1-00:00:00 --partition=general --ntasks=1 --pty /bin/bash -l
module load anaconda

#This script will download a new qiime2 environment
conda env create \
  --name qiime2-amplicon-2025.10 \
```

```
  --file
https://raw.githubusercontent.com/qiime2/distributions/refs/heads/dev/2025.10/amplicon/released/qiime2-amplicon-ubuntu-latest-conda.yml

#Confirm that the new Qiime2 environment works:
source activate qiime2-amplicon-2025.10

#After successfully downloading the new version (and confirming it works), delete the old version. For me, this was version 2024.5:
conda remove --name qiime2-amplicon-2024.5 --all
```